

CS-302: Artificial Intelligence

LECTURE NO 8:

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Organization of Lecture

1. Informed Search Strategies (*Continuation of Lecture 6&7*)
 1. Hill-Climbing Search
 2. Branch and Bound Technique
 3. Generate and Search Technique in A.I
 4. State Space Search

1. Hill-Climbing Search (1/5)

1) Hill - Climbing Search

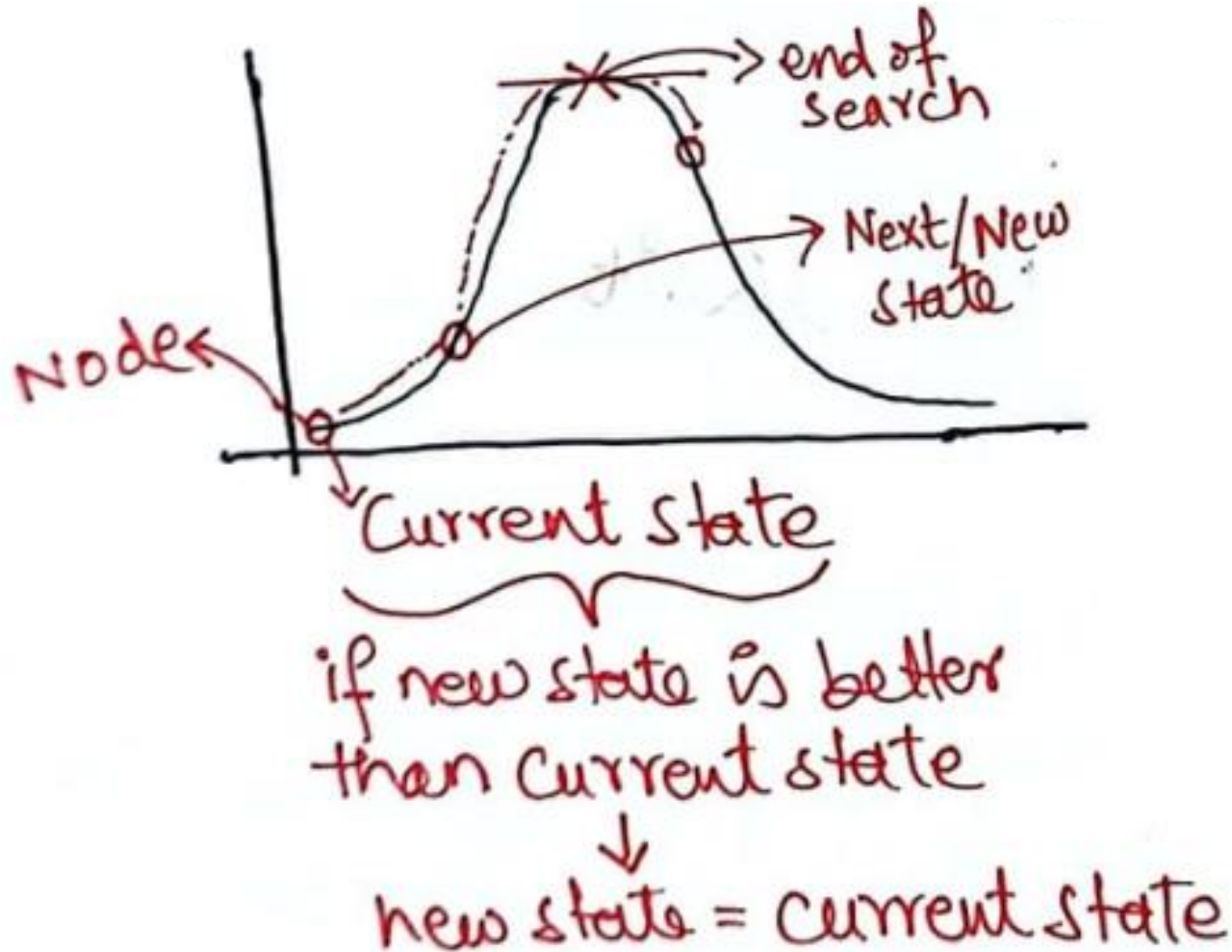
↳ Variant of generate and test method in which feedback from test procedure is used to help generator decide which direction to move in search space.] $\Rightarrow$ Always moves in a single direction.

↳ It is like DFS (Depth First Search).

↳ '3' Important points

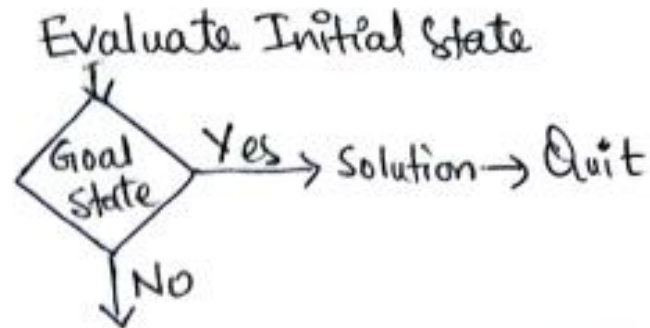
- i - Local Search Algorithm (Local Maxima).
- ii - It uses Greedy Approach.
- iii - No Back Tracking.

1. Hill-Climbing Search (2/5)

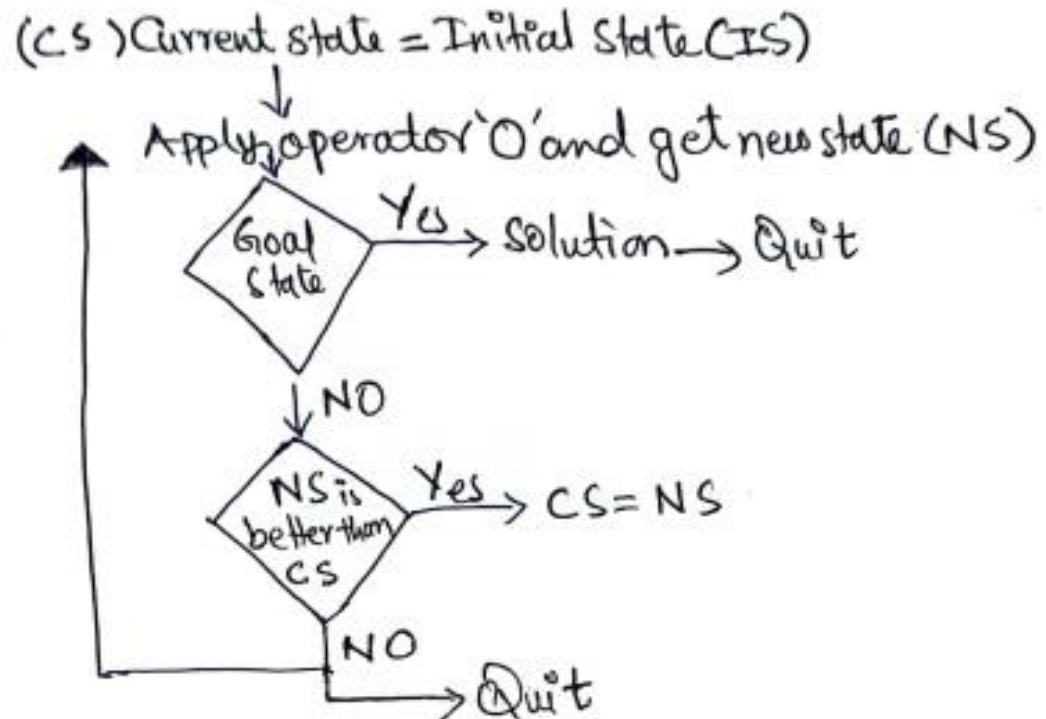


1. Hill-Climbing Search (3/5)

Flowchart:



Operator 'O' = New Node



1. Hill-Climbing Search (4/5)

Example

1	2	4
5		7
3	6	8

⇒ Starting State/Initial State

⇒ Let's say the next three states have heuristic values as 4, 5 & 6. The state with minimum heuristic value will be selected as Next/New State.

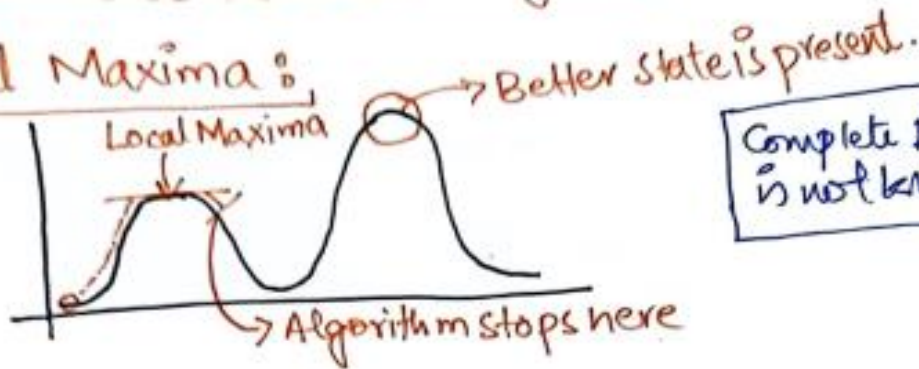
(4) (5) ~~(6)~~

↳ We will choose '4' & discard '5' & '6'

1. Hill-Climbing Search (5/5)

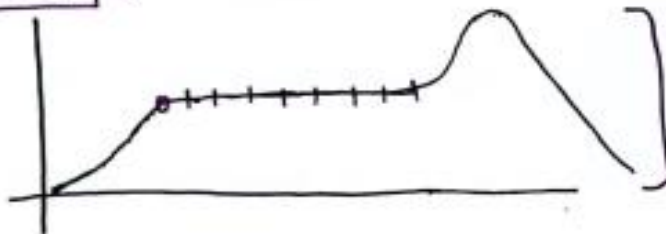
Limitations of Hill-Climbing Search:

① Local Maxima:



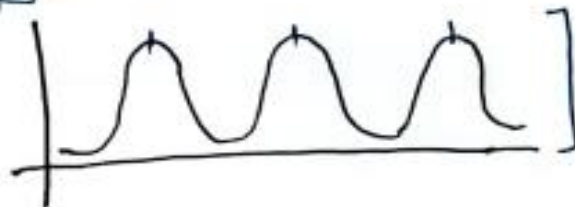
Complete Domain knowledge is not known.

② Plateau:



Algo gets same value / same solⁿ for some time.

③ Ridge:



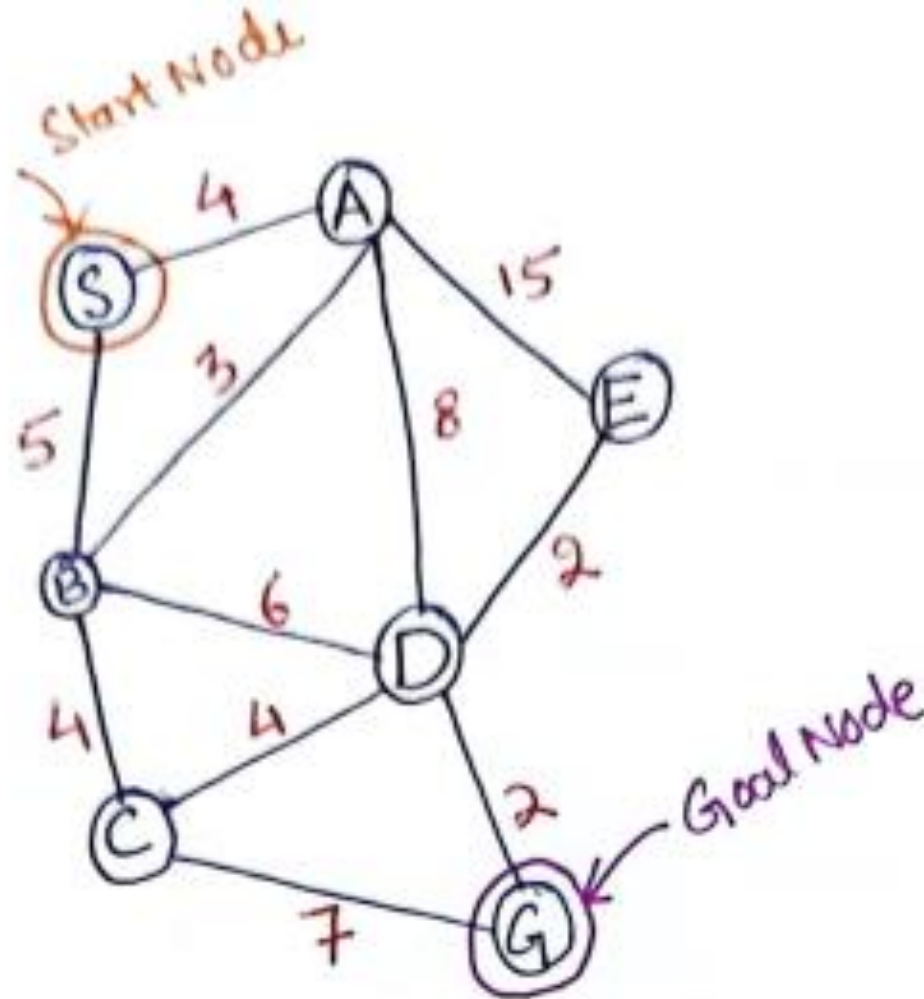
All successors get same value or same height.

2. Branch and Bound Technique (1/4)

- ↳ Branch and Bound Search is a way to combine space saving of depth first search with heuristic information.
- ↳ Optimal solution is chosen among many solution paths.
- ↳ Idea in Branch-and-Bound search is to maintain lowest-cost path to a GOAL found so far, and its cost.
- ↳ Typically used with depth-first search.
- ↳ No back tracking [i.e., only forward direction].
- ↳ Two parts
 - i- Branch: Several choices are found
 - ii- Bound: setting Bound on solution Quality
 - ↳ Pruning: Trimming of branches where solution Quality is poor.

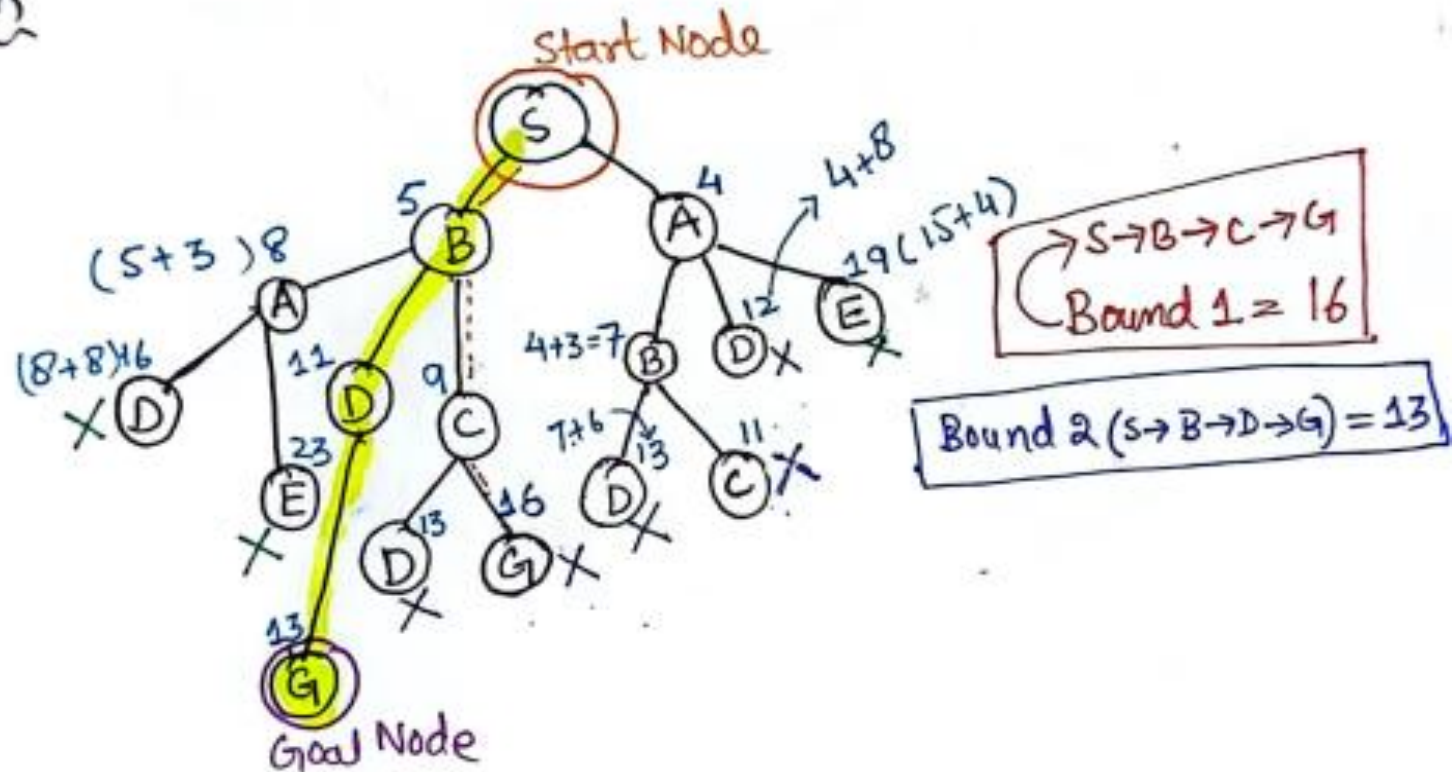
2. Branch and Bound Technique (2/4)

Example:

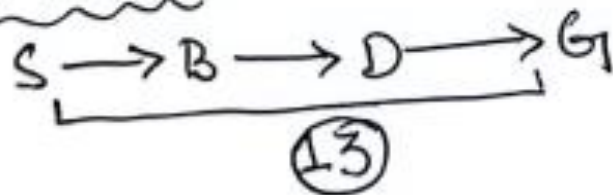


2. Branch and Bound Technique (3/4)

Solution



Optimal Path



2. Branch and Bound Technique (4/4)

Disadvantages:

- i- load balancing factor of branch and bound makes the parallelization difficult.
- ii- It is limited to small sized network.

3. Generate and Search Technique in A.I (1/3)

3) Generate and Test Search in AI:

- ↳ In this technique all the solutions are generated & tested for best solution.
- ↳ It ensures that the best solution is checked from all possible generated solutions.
- ↳ Also known as British Museum Search Algorithm.
- ↳ Heuristic ^{works like} DFS with back tracking
- ↳ If the solution is found, Quit.

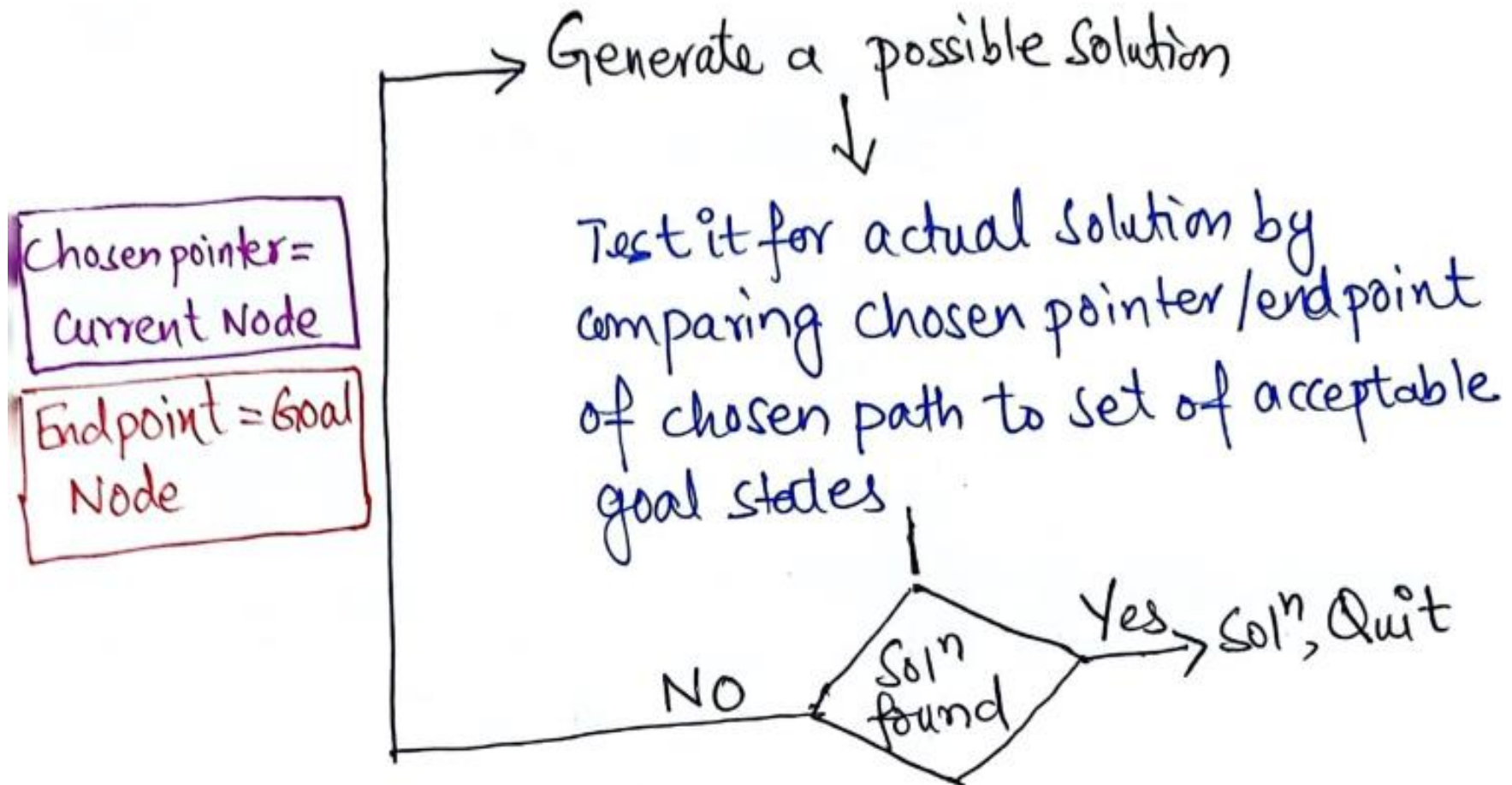
3. Generate and Search Technique in A.I (2/3)

Properties of Good Generators:

- ① Complete (Always produces a good solution).
- ② Non-Redundant
- ③ Informed (should be an informed Search Technique).

3. Generate and Search Technique in A.I (3/3)

Flow Diagram of Generate and Test Search:



4. State Space Search (1/5)

4) State Space Search:

- ↳ It is used in problem solving.
- ↳ It is a process used in A.I in which successive configurations or states of an instance are considered with intention of finding a GOAL state with desired property.
- ↳ Problems are modelled as State Space
set of states in which a problem can exist.

4. State Space Search (2/5)

↳ Representation:

$S: [S, A, \text{Action}(S), \text{Result}(s, a), \text{Cost}(s, a)]$

Where

- S = set of all possible states of a Problem
- A = set of all possible actions
- $\text{Action}(S)$ = It is a function which determines which action is possible for current state.
- $\text{Result}(s, a)$ = It is a function which tells about the next state reached by performing action 'a' on a current state 's'.
- $\text{Cost}(s, a)$ = It is a function which tells about the cost when action 'a' is performed on state 's'.

4. State Space Search (3/5)

Example 1 Eight-Tile Puzzle Problem

Start State (S)

1	4	3
2	///	5
8	6	7

GOAL STATE (G)

1	2	3
8	///	4
7	6	5

Possible Actions: 4  } Legal Moves for a particular state

↳ for example 'Right Move' on Start State (S) above :

1	4	3
2	5	///
8	6	7

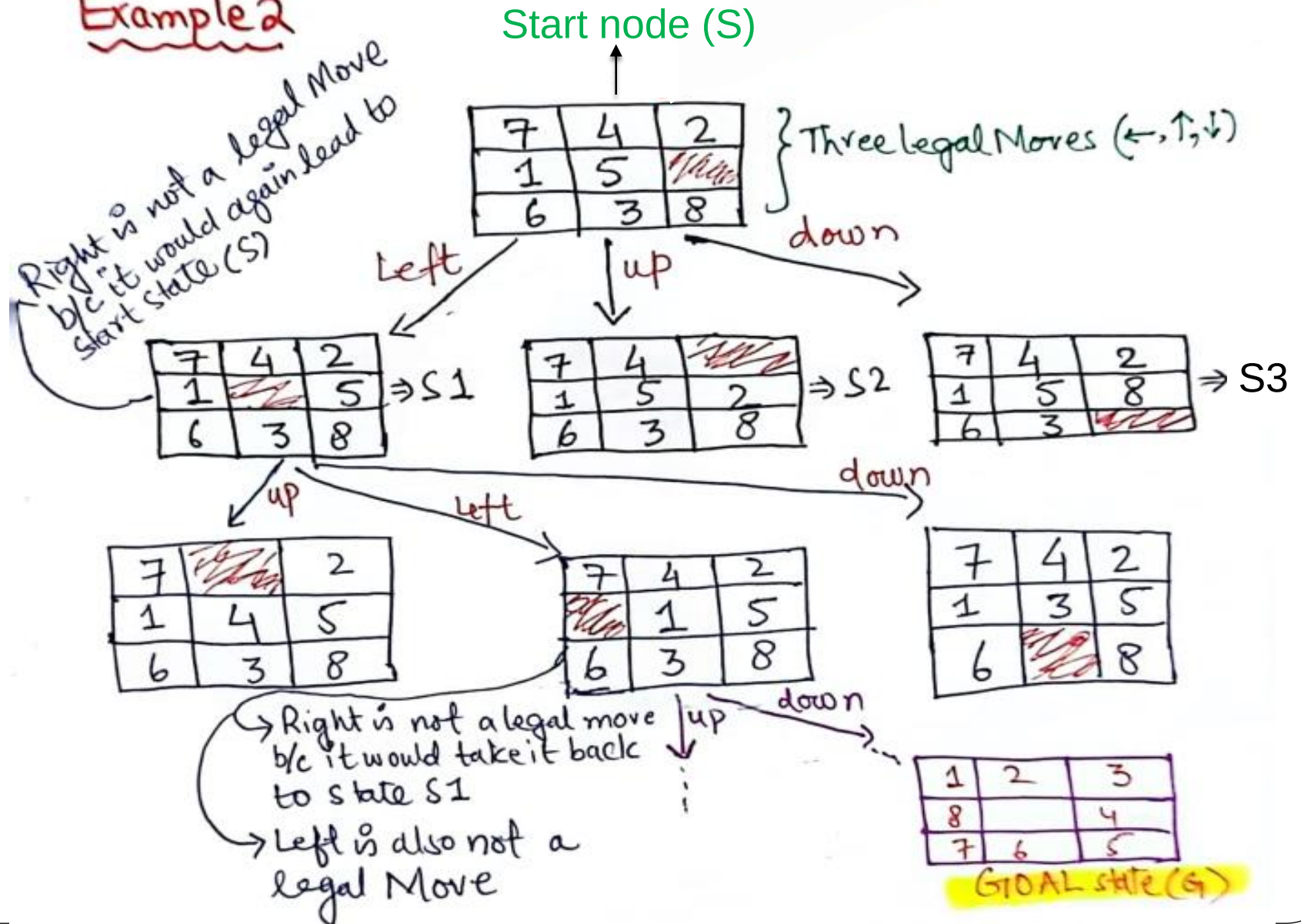
↓
New State (S1)

} Right is not possible now } illegal Move

 } Legal Moves

4. State Space Search (4/5)

Example 2



4. State Space Search (5/5)

Home Task:

Determine the minimum number of moves to reach to GOAL node (G) in Example 2.

References: Textbooks

1. *“Artificial Intelligence”, by Elaine Rich and Kevin Knight, (2006), McGraw Hill companies Inc., Chapter 1-22 page 1-613.*
2. *“Artificial Intelligence: A Modern Approach” by Stuart Russell and Peter Norvig, (2010), Prentice Hall, Chapter 1-27. page 1-1151.*
3. *“Computational Intelligence: A logical Approach”, by David Poole, Alan Mackworth, and Randy Goebel, (1998), Oxford University Press, Chapter 1-12, page 1-608.*
4. *“Artificial Intelligence: Structures and Strategies for Complex Problem Solving”, by George F.Lugar, (2002), Addison-Wesley, Chapter 1-16, page 1-743.*
5. *“AI: A new Synthesis”, by Nils J.Nilsson, (1998), Morgan Kaufmann Inc., Chapter 1-25, Page 1-493.*
6. *“Artificial Intelligence: Theory and Practice” by Thomas Dean, (1994), Addison-Wesley, Chapter 1-10, Page 1-650.*
7. *Related documents from open source, mainly internet.*